a. JOEL THEO C. GALLARDE
b. Library Management System
c. The Library Management System is a console-based application developed in Assembly Language designed to digitize the tracking of books in a small library. It features a secure login panel allowing multiple students to register and access the system.

   Once logged in, the user can manage the library inventory through a main menu. Capabilities include adding new books to specific shelf positions (1-20), viewing the complete catalog in a tabular format, updating existing book details (Author and Title), removing books from the system (with automatic array shifting), and checking the current capacity of the library shelves. The system handles input validation to prevent duplicate titles and ensures shelf limits are not exceeded.

d. **Authentication Inputs:** Username, Password, Full Name.

   Menu Selection: Single character inputs (Keys '1' through '6').

   Location Input: Shelf Position (Numeric string, range 1-20).

   Book Data: Author Name (up to 20 chars), Book Title (up to 40 chars).

   Search/Identifier: Book ID (e.g., BOOK-05) for updating or removing items.

   System Controls: "Press any key" to continue.

e. **User Registration:** Checks if a username exists; if unique, stores credentials and full name in memory arrays.

   User Login: Validates entered username and password against stored records.

   Add Book: Validates shelf position (checks range and occupancy), checks forduplicate titles, generates a unique Book ID, and stores data.

   View Catalog: Iterates through the book arrays and displays all items in a formatted table.

   Update Book: Searches for a book by ID; if found, overwrites the Author and Title with new inputs.

   Remove Book: Searches for a book by ID; if found, deletes it and shifts all subsequent array elements to fill the gap.

   Check Capacity: Calculates and displays the number of empty slots vs. occupied slots.

 Validation Utilities: Functions to compare strings, convert strings to numbers, and clear the screen.

f. **Status Messages:** "SUCCESS: Operation completed", "ERROR: Invalid username", "Library shelves are full".

 Data Displays: The "Library Catalog" table showing Position, ID, Author, and Title.

 Statistics: "Empty Shelf Positions: [X]", "Books in Stock: [Y]".

 Interface Elements: Main Menu options, Login Panel, Header banners.

g. 1. **Initialization:** Load data segment and clear screen.

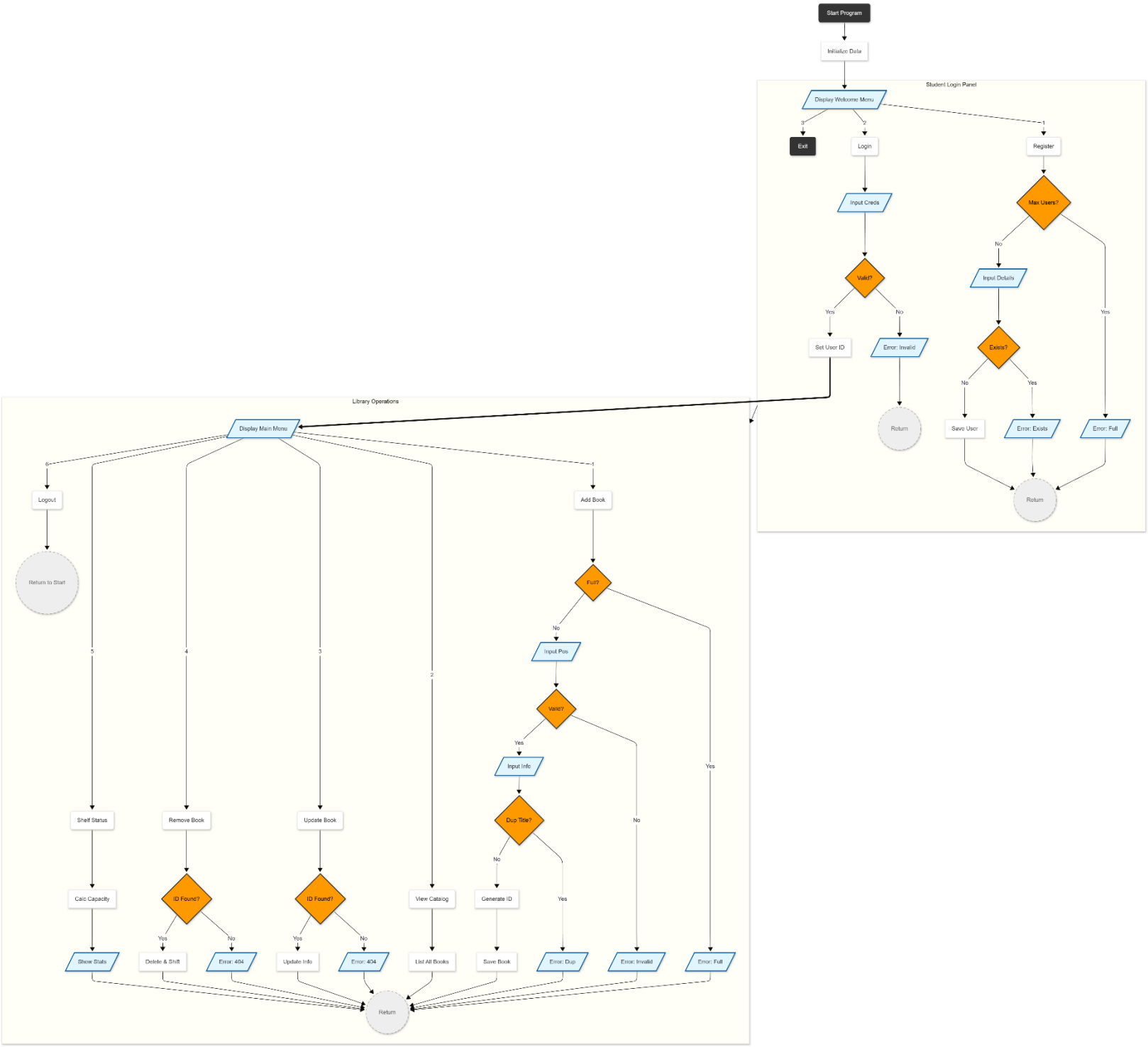
2. **Welcome Loop (Start):** Display "Student Login Panel".

- If user selects **1 (Register)**: Prompt for new Username, Password, Name. Store in memory. Return to start.
- If user selects **2 (Login)**: Prompt for credentials. Compare with stored data. If match, proceed to Main Menu. If fail, show error.
- If user selects **3 (Exit)**: Terminate program.

3. **Main Menu Loop:** Display options 1-6.

- **Option 1 (Add):** Prompt for Shelf Location (1-20). Validate availability. Prompt for Author/Title. Save to arrays.
- **Option 2 (View):** Loop through BOOK_COUNT. Print header and row data for every book.
- **Option 3 (Update):** Ask for Book ID. Search arrays. If found, prompt for new details and overwrite.
- **Option 4 (Remove):** Ask for Book ID. Search arrays. If found, shift all following bytes up by one index to delete.
- **Option 5 (Status):** Calculate MAX_BOOKS - BOOK_COUNT. Display result.
- **Option 6 (Logout):** Jump back to Welcome Loop.

h. Overall Flowchart drawing



i. USER INTERFACE draft illustrations/drawings.

Screen 1: Login Panel

Plaintext

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LIBRARY MANAGEMENT SYSTEM
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STUDENT LOGIN PANEL
1. Register New Student
2. Login
3. Exit System

Enter choice: _
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Screen 2: Main Menu

Plaintext

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LIBRARY MANAGEMENT SYSTEM
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MAIN MENU
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1. Add New Book to Shelf
2. View Library Catalog
3. Update Book Details
4. Remove Book from Shelf
5. View Shelf Capacity
6. Logout
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Screen 3: View Catalog

Plaintext

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LIBRARY CATALOG

POS   ID          AUTHOR (20)           TITLE (40)
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01    BOOK-01     J.K. Rowling          Harry Potter
05    BOOK-05     J.R.R. Tolkien        The Hobbit

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Press any key to continue...
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